

Get Ready to CRUSH Your DevNet Associate Exam



DEVASC (200-901) Practice Exam

CREATED BY
Kevin Wallace Training, LLC

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We at Kevin Wallace Training, LLC (<https://kwtrain.com>) want to help you earn your certification, not memorize brain dumps.

So, all questions in this practice exam were entirely created by us based on Cisco's DEVASC (200-901) exam blueprint.

Even more important that the questions though are our **detailed explanations**, found in the second portion of this document.

We want you to know why each correct answer is correct. So, please spend plenty of time reading through the explanations for each question.

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I wish you all the best in your studies,

Kevin Wallace, CCIEx2 #7945 Emeritus (Enterprise Infrastructure & Collaboration)

DevNet Associate (DEVASC 200-901)

Practice Exam – Questions

1. Which of the following is a characteristic of XML format?
 - A. Pre-defined tags
 - B. Concerned with carrying data
 - C. Used for displaying data
 - D. Key:value pairs
2. Which of the following is NOT a characteristic of JSON format?
 - A. Comments are not allowed
 - B. Whitespace has no impact on format
 - C. No ability to use arrays
 - D. Uses a bracketing system to identify objects
3. Which data format is used by Cisco Modeling Labs (CML) for importing and exporting lab files?
 - A. XML
 - B. IOSXE
 - C. JSON
 - D. YAML
4. Which built in Python library can be used to parse XML data?
 - A. ElementTree
 - B. XMLTree
 - C. Tree.getrroot
 - D. Print(root)
5. Reference the following abbreviated snippet from a JSON file:

```
{"device":{"hostname":"R1","interface":[{"GigabitEthernet". . .
```

What will be the returned output when parsing through the data using the built in JSON library in Python ending in the following print statement: `print(data['device']['hostname'])`

- A. hostname
- B. device
- C. interface
- D. R1

6. Reference the following abbreviated snippet from a YAML file:

```
OFFICE:
  device:
    hostname: R1
    interface:
      GigabitEthernet:
        . . .
```

Which of the following print statements can be used to return the value “R1” when parsing through the YAML data in Python?

- A. `print(data['OFFICE'])`
- B. `print(data['OFFICE']['device']['hostname'])`
- C. `print(data['hostname'])`
- D. `print(data['device']['hostname'])`

7. What is the initial phase used when employing a Test-Driven Development methodology?

- A. Coding
- B. Planning
- C. Testing
- D. Debugging

8. What is the term for our general plan at the beginning of development when using the Agile Method?

- A. Sprint
- B. Product backlog
- C. Cycle
- D. Scrum

9. What is the final step in the Lean Method before the cycle repeats?
- A. Seek perfection
 - B. Create flow
 - C. Map value stream
 - D. Establish pull
10. Which of the following is a characteristic of the Waterfall Method?
- A. Customer feedback is timely and constant
 - B. Value is created throughout the process
 - C. Much more inexpensive than other methods
 - D. Everything is planned at the beginning of a project
11. Which Python structure allows us to run the same piece of code multiple times in a program to repurpose the code?
- A. Classes
 - B. Modules
 - C. Functions
 - D. Blocks
12. Which Python structure is closely associated with objects?
- A. Modules
 - B. Classes
 - C. Functions
 - D. Blocks
13. Which Python structure allows us to import code from other external files into our main code?
- A. Blocks
 - B. Functions
 - C. Modules
 - D. Classes
14. Which common design method leverages the Separation of Concerns (SoC) principle for designing modular code?

- A. Model-View-Controller (MVC)
- B. Observer
- C. Version Control
- D. Git

15. Which design method is associated with one-to-many relationships, such as database design?

- A. Version Control
- B. Model-View-Controller (MVC)
- C. Logging
- D. Observer

16. Which principle means we are tracking all the changes made to a file or a set of files in a historical perspective?

- A. Observer
- B. Version Control
- C. Model-View-Controller (MVC)
- D. Logging

17. Which Git version control operation allows us to create a copy of a repository on our local file system?

- A. Commit
- B. Clone
- C. Push
- D. Merge

18. Which Git version control operation allows us to track the status of a file including any changes made?

- A. Add/Remove
- B. Diff
- C. Branch
- D. Clone

19. Which Git version control operation moves a file from our index (or staging area) to our local repository?
- A. Merge
 - B. Push
 - C. Commit
 - D. Push/Pull
20. Which Git version control operation will add all tracked files from our local file system to a central remote repository such as GitHub?
- A. Commit
 - B. Push/Pull
 - C. Clone
 - D. Diff
21. Which Git version control operation allows us a safe way to separate our development area from a main branch repository containing known working code?
- A. Clone
 - B. Add/Remove
 - C. Diff
 - D. Branch
22. Which Git version control operation allows us to join the contents of a branch development area with a main repository locally?
- A. Diff
 - B. Push/Pull
 - C. Merge
 - D. Add/Remove
23. Which Git version control operation is used as a file comparison tool?
- A. Merge
 - B. Branch
 - C. Clone
 - D. Diff

24. Which method of authentication for REST API is typically limited only to operations where the READ function is required?
- A. Basic Authentication
 - B. OAUTH Authentication
 - C. Open Authentication
 - D. API Key Authentication
25. What type of HTTP messages are used by Webhooks?
- A. HTTP UPDATE
 - B. HTTP READ
 - C. HTTP POST
 - D. HTTP DELETE
26. Which of the following is NOT a characteristic of a true RESTful API?
- A. Stateful
 - B. Uniform Interface
 - C. Cacheable or Non-cacheable Label
 - D. Layered System
27. When using Python to make a REST API call, which built in library can be used in the script to perform a GET request?
- A. Requests
 - B. Get
 - C. Math
 - D. XML
28. Which range of HTTP response codes are used to indicate a client-side error?
- A. 2XX codes
 - B. 3XX codes
 - C. 4XX codes
 - D. 5XX codes
29. In which part of an HTTP response would we see things like the host, user agent and cookie?

- A. URL
- B. Response code
- C. Arguments
- D. Headers

30. Which HTTP response code indicates a likely issue with incorrect authentication credentials?

- A. 500 Internal Server Error
- B. 403 Forbidden
- C. 401 Unauthorized
- D. 501 Not Implemented

31. Which API authentication mechanism is the least secure?

- A. OAUTH Authentication
- B. Basic Authentication
- C. Token Authentication
- D. API Key Authentication

32. Which API authentication mechanism allows us to enter our username and password a single time to generate an encrypted value for accessing resources?

- A. Token Authentication
- B. API Key Authentication
- C. Basic Authentication
- D. Secure Authentication

33. Which API authentication mechanism is primarily used for overall project access rather than at the individual user level?

- A. General Authentication
- B. Encrypted Authentication
- C. Token Authentication
- D. API Key Authentication

34. Which subset of common HTTP methods are used by Remote Procedure Calls (RPCs)?

- A. POST and PUT
- B. PATCH and DELETE
- C. GET and POST
- D. PUT and GET

35. Which style of API does not use real-time communication?

- A. Synchronous
- B. Asynchronous
- C. One-way
- D. Delayed

36. What is the common name for a set of tools, libraries and documentation that allows us to interact with a REST API?

- A. Framework
- B. Program
- C. Script
- D. SDK

37. Which of the following is appropriate for obtaining a list of the organizations available through the GET method when using Postman to interact with the Cisco Meraki API?

- A. `{{baseUrl}}/organizations/:organizationId`
- B. `{{baseUrl}}/organizations`
- C. `{{baseUrl}}/organizations/:organizationId/networks`
- D. `{{baseUrl}}/organizations/:organizationId/devices`

38. Which type of API authentication method is used by Cisco DNA Center?

- A. Token Authentication
- B. Basic Authentication
- C. API Key Authentication
- D. Open Authentication

39. Which type of API authentication method is used by Cisco APIC?

- A. Open Authentication
- B. API Key Authentication

- C. Basic Authentication
- D. Token Authentication

40. Which area in Postman allows you to create preconfigured areas to be used with specific devices, which can hold key:value pair values for variables like the base URL, the token, the username, and the password?

- A. Environments
- B. Flows
- C. Collections
- D. Requests

41. With Cisco Network Services Orchestrator, what are required to translate between XML format and our network configuration?

- A. YAML
- B. JSON
- C. NEDS
- D. LIB

42. Within a Cisco UCS-based network, what type of device is used to interconnect all UCS-capable servers into a single point of management?

- A. UCS Fabric Extender
- B. UCS Series Servers
- C. UCS Fabric Interconnects
- D. UCS Manager

43. Which component used within Cisco UCS Director are the largest units of information?

- A. Workflows
- B. Tasks
- C. Volumes
- D. Libraries

44. Which area within Cisco Intersight shows all the devices under our control which we have claimed?

- A. Assets

- B. Profiles
- C. Targets
- D. Policies

45. Which method is required for creating a new room in Cisco Webex through API interaction?

- A. POST
- B. PUT
- C. GET
- D. UPDATE

46. Which type of API interface is used by Cisco collaboration endpoint software such as Cisco CE and Cisco RoomOS?

- A. ceAPI
- B. cAPI
- C. osAPI
- D. xAPI

47. Which type of API interface is used for Cisco CUCM interaction?

- A. Open API
- B. xAPI
- C. AXL API
- D. SDK API

48. Which Cisco collaboration solution provides an agent and supervisor desktop that we can affect through REST API interaction?

- A. Finesse
- B. Cisco UCM
- C. Cisco Webex
- D. Cisco UCS

49. Which key:value pair used to identify the authentication token with the Firepower API?

- A. X-auth-token:<authentication token value>
- B. X-auth-access-token:<authentication token value>

- C. X-auth-bearer-token:<authentication token value>
- D. X-auth-secure-token:<authentication token value>

50. Which cloud-based Cisco platform is used as a secure internet gateway for things like malware prevention and secure DNS routing?

- A. AMP
- B. Umbrella
- C. ISE
- D. ThreatGrid

51. When interacting with Cisco Secure Endpoint (formerly Cisco AMP) through API, what type of encoding should we use for the client ID and API key for use within Postman?

- A. Unicode
- B. ASCII
- C. Base64
- D. XML

52. Which appropriate key:value pair will set our content type to be formatted in JSON when interacting with the Cisco ISE API?

- A. Content-type:application/json
- B. Content:type/json
- C. Content:json
- D. Content-type:json

53. Which type of malware analysis used by Cisco Secure Malware Analytics (formerly Cisco ThreatGrid) is performed in an isolated virtual environment?

- A. Dynamic malware analysis
- B. Static malware analysis
- C. Virtual malware analysis
- D. Isolated malware analysis

54. Which command within the Cisco NX-OS CLI line will enable the developer sandbox?

- A. feature nsox
- B. feature sandbox
- C. feature nxapi

D. feature devbox

55. Which DevNet resource provides a way for us to instantly access Cisco devices through a web browser?

- A. API Documentation
- B. Learning Labs
- C. Code Exchange
- D. Sandbox

56. Which DevNet resource provides an online, publicly available repository of code related to Cisco technologies?

- A. API Documentation
- B. Learning Labs
- C. Code Exchange
- D. Sandbox

57. What is the recommended way to obtain in-depth assistance with topics related to DevOps and Cisco DevNet principles from the DevNet Developer Support site?

- A. Knowledge Base
- B. Community Forum
- C. Chat
- D. Support Ticket

58. What is the term for attributes outlined such as integers or strings within the YANG data model?

- A. Key
- B. Namespace
- C. Leaf
- D. List

59. When a router receives a NETCONF request from another device, what is the term for this device within the context of NETCONF?

- A. Syslog
- B. Trap

- C. Manager
- D. Agent

60. Which protocol is used by RESTCONF for sending and receiving information?

- A. SNMP
- B. SSH
- C. HTTPS
- D. DNS

61. Which type of cloud service model provides users with quick access to cloud-based web applications?

- A. SaaS
- B. PaaS
- C. IaaS
- D. DaaS

62. Which type of cloud deployment model is provisioned for open use?

- A. Private Cloud
- B. Community Cloud
- C. Public Cloud
- D. Hybrid Cloud

63. Within the NIST Cloud Deployment Reference Architecture, under which role would we categorize end users of a cloud solution?

- A. Cloud Provider
- B. Cloud Broker
- C. Cloud Consumer
- D. Cloud Carrier

64. Which of the following is NOT a characteristic of Edge Computing?

- A. Better security and privacy
- B. Slightly higher cost associated
- C. Lower latency for real-time data
- D. Balance between local and cloud resources

65. Which type of Hypervisor runs directly on a server's hardware?
- A. Type-1
 - B. Type-2
 - C. Type-3
 - D. Type-4
66. Which of the following is NOT a characteristic of Containers?
- A. Run on a single host OS
 - B. Smaller than VMs
 - C. Consist of packages of applications, software, and libraries
 - D. More resource intensive than VMs
67. Under which cloud service model would we categorize Serverless Computing?
- A. IaaS
 - B. SaaS
 - C. FaaS
 - D. PaaS
68. Which step within the CI/CD pipeline involves staging and deploying the application?
- A. Source
 - B. Deploy
 - C. Build
 - D. Test
69. After writing a Unit Test, what is next second step which will inform the remainder of our Test-Drive Development cycle?
- A. Testing fails
 - B. Refactoring
 - C. Testing passes
 - D. Merging code

70. What is the term for a text document containing commands used for assembling images in Docker?
- A. Dockercontainer
 - B. Dockerfile
 - C. Dockerimage
 - D. Dockerscript
71. At which layer of the Vulnerability Stack for threats would we find web applications?
- A. Network
 - B. Operating system
 - C. Databases
 - D. Third-party components
72. According to the most recent OWASP Top 10 Web Application Threats list, what is rated as the number one danger to our web applications?
- A. Injection
 - B. Cryptographic failures
 - C. Broken access control
 - D. Insecure design
73. Which of the following is not an advantage of Model-Driven Programmability for infrastructure automation?
- A. Does not require vendor-specific syntax
 - B. Unified way of interacting
 - C. Complete control over devices
 - D. Reliable and repeatable management and configuration
74. With controller-level management, which type of interface is used for integrating third-party systems through REST APIs?
- A. Northbound
 - B. Southbound
 - C. Eastbound
 - D. Westbound

75. Which type of network management approach features a single central point of administration?
- A. Central-level management
 - B. Server-level management
 - C. Device-level management
 - D. Controller-level management
76. What is the name of the YAML file in pyATS where we define devices to run tests against?
- A. Inventory.yaml
 - B. Assets.yaml
 - C. Testbed.yaml
 - D. Devices.yaml
77. Which approach to Infrastructure as Code (IaC) involves defining what the desired end state of the system should look like?
- A. Manual approach
 - B. Defining approach
 - C. Imperative approach
 - D. Declarative approach
78. Which well-known automation tool has structures such as the Bookshelf and Test Kitchen?
- A. Chef
 - B. Puppet
 - C. Ansible
 - D. Python
79. What is the term used with Puppet automation for a description of a desired machine state?
- A. Module
 - B. Manifest
 - C. Resource
 - D. Collection

80. Which Ansible structure contains a list of nodes which are under the management of the Ansible Controller?
- A. Playbook
 - B. Inventory File
 - C. Module
 - D. Play
81. When interpreting a YANG data model, what indicates that a particular field is configurable, as opposed to only being able to retrieve information about the field?
- A. ro
 - B. rw
 - C. conf
 - D. string
82. What port is used for NETCONF by Cisco IOS-XE?
- A. Port 443
 - B. Port 9339
 - C. Port 50052
 - D. Port 830
83. What port is used for RESTCONF by Cisco IOS-XE?
- A. Port 443
 - B. Port 9339
 - C. Port 50052
 - D. Port 830
84. You have two Git branches on your local system, one named "main" and the other named "newfeatures." When working in the NewFeatures branch, which terminal command will allow you to see the differences between these branches and files?
- A. git diff main
 - B. git diff newfeatures
 - C. git diff branch
 - D. git diff branch -list

85. Which of the following is NOT a best practice recommendation for code review?
- A. Review less than 400 lines at a time
 - B. Take breaks throughout code review
 - C. Review all code in one sitting
 - D. Review at a pace no faster than 400 lines per hour
86. Which of the following is NOT a characteristic of Sequence Diagrams?
- A. Used to illustrate REST APIs visually
 - B. Provide the necessary API keys for interaction
 - C. Common method for developer teams
 - D. Helps outline project goals and requirements
87. What is the size of a MAC address?
- A. 64 bits
 - B. 32 bits
 - C. 128 bits
 - D. 48 bits
88. Which interface mode is used to carry traffic for multiple designated VLANs?
- A. VLAN mode
 - B. Access mode
 - C. Trunk mode
 - D. Multi mode
89. What is the size of an IPv4 address?
- A. 64 bits
 - B. 32 bits
 - C. 128 bits
 - D. 48 bits
90. Which address structure is used by a router for sending traffic to its next destination?
- A. IP address
 - B. MAC address

- C. DNS address
- D. HTTP address

91. At which layer of the OSI Model do we find the packets?

- A. Layer 1
- B. Layer 2
- C. Layer 3
- D. Layer 4

92. At which layer of the OSI model do we categorize a typical switch?

- A. Layer 1
- B. Layer 2
- C. Layer 3
- D. Layer 4

93. What type of device is necessary for routing traffic between different VLANs?

- A. Layer 4 device
- B. Layer 2 device
- C. Layer 1 device
- D. Layer 3 device

94. What is the term for the firewall zone connecting out to the public Internet?

- A. Outside
- B. Inside
- C. DMZ
- D. Demarcation

95. Which network device is ideally suited for providing redundancy and low latency?

- A. Load balancer
- B. Firewall
- C. Virtual server
- D. Next-generation router

96. Which plane of operation in networking is where we find user traffic transiting the router, not destined to, or initiated by the router itself?
- A. Data Plane
 - B. Management Plane
 - C. Control Plane
 - D. Traffic Plane
97. Which plane of operation in networking is considered to have its own ingress and egress ports?
- A. Data Plane
 - B. Management Plane
 - C. Control Plane
 - D. Traffic Plane
98. Which plane of operation in networking is where we would see protocols like SSH and Telnet being used?
- A. Data Plane
 - B. Management Plane
 - C. Control Plane
 - D. Traffic Plane
99. Which step during the DHCP exchange uses a broadcast message to locate a DHCP server?
- A. Offer
 - B. Request
 - C. Discover
 - D. Acknowledgement
100. Which well-known port is used by Domain Name System (DNS)?
- A. 80
 - B. 1400
 - C. 53
 - D. 443

101. When using Network Address Translation, what is the term for addresses found inside the NAT-enabled network?

- A. Inside Local
- B. Inside Global
- C. Outside Local
- D. Outside Global

102. Which version of SNMP added additional security features for encryption, integrity checking, and authentication services?

- A. SNMPv1
- B. SNMPv2
- C. SNMPv2c
- D. SNMPv3

103. Which well-known port is used by Network Time Protocol (NTP)?

- A. TCP 123
- B. UDP 123
- C. UDP 189
- D. TCP 189

104. What is the second step in the TCP 3-way handshake process?

- A. SYN-ACK
- B. SYN
- C. ACK
- D. ACK-SYN

105. What range of ports is referred to as the well-known port range?

- A. 49151 - 65535
- B. 1024 - 49151
- C. 0 - 1023
- D. 500 - 1024

106. At which OSI Model layer would we perform troubleshooting if we have a misconfigured or disabled switch port?

- A. Network Layer
- B. Data Link Layer
- C. Physical Layer
- D. Transport Layer

107. At which OSI Model layer are we performing troubleshooting when using Cisco IOS commands such as “show mac address-table”?

- A. Network Layer
- B. Data Link Layer
- C. Physical Layer
- D. Transport Layer

108. At which OSI Model layer are we performing troubleshooting when using the ping command utility?

- A. Network Layer
- B. Data Link Layer
- C. Physical Layer
- D. Transport Layer

109. At which OSI Model layer are we performing troubleshooting when checking firewall access control lists for issues?

- A. Network Layer
- B. Data Link Layer
- C. Physical Layer
- D. Transport Layer

DevNet Associate (DEVASC 200-901)

Practice Exam – Answers

1. Which of the following is a characteristic of XML format?

- A. Pre-defined tags
- B. Concerned with carrying data
- C. Used for displaying data
- D. Key:value pairs

Answer: B

Explanation: XML format is primarily concerned with carrying data. This is opposed to other examples like HTML, which is concerned with displaying the data. Pre-defined tags are another characteristic of HTML, whereas XML has no pre-defined tags. XML format does not use key:value pairs as JSON and YAML formats do.

Video Reference: 1.1.1 XML Format

2. Which of the following is NOT a characteristic of JSON format?

- A. Comments are not allowed
- B. Whitespace has no impact on data
- C. No ability to use arrays
- D. Uses a bracketing system to identify objects

Answer: C

Explanation: JSON format does have the ability to use arrays, which makes it unique from XML. Other characteristics include comments not being supported in the data, the ability to insert whitespace with no impact on the data, and the use of a bracketing system to identify objects and arrays.

Video Reference: 1.1.2 JSON Format

3. Which data format is used by Cisco Modeling Labs (CML) for importing and exporting lab files?

- A. XML
- B. IOSXE
- C. JSON

D. YAML

Answer: D

Explanation: YAML is used within CML for importing and exporting lab files. YAML uses a hierarchical format as other data formats do and leverages key:value pairs like JSON. This is one of the most human-readable data formats for networking.

Video Reference: 1.1.3 YAML Format

4. Which built in Python library can be used to parse XML data?

- A. ElementTree
- B. XMLTree
- C. Tree.getrroot
- D. Print(root)

Answer: A

Explanation: Parsing means we are converting a certain set of data into another format. When parsing XML using Python, the built in ElementTree library has functionality for reading and manipulating XML data.

Video Reference: 1.2.1 XML Parsing

5. Reference the following abbreviated snippet from a JSON file:

```
{“device”: {“hostname”: “R1”, “interface”: [{“GigabitEthernet”. . .
```

What will be the returned output when parsing through the data using the built in JSON library in Python ending in the following print statement: *print(data[‘device’][‘hostname’])*

- A. hostname
- B. device
- C. interface
- D. R1

Answer: D

Explanation: The given print statement will index through the shown JSON snippet by referencing key:value pairs. This print statement references the “hostname” key and returns the associated value for this key. In this JSON file, the value associated with the hostname key is “R1”.

Video Reference: 1.2.2 JSON Parsing

6. Reference the following abbreviated snippet from a YAML file:

```
OFFICE:
  device:
    hostname: R1
    interface:
      GigabitEthernet:
. . .
```

Which of the following print statements can be used to return the value “R1” when parsing through the YAML data in Python?

- A. `print(data['OFFICE'])`
- B. `print(data['OFFICE']['device']['hostname'])`
- C. `print(data['hostname'])`
- D. `print(data['device']['hostname'])`

Answer: B

Explanation: The print statement `print(data['OFFICE']['device']['hostname'])` will iterate through the YAML document and specifically return the value for the “hostname” key in the document. In this case, that value is “R1”, as seen in the abbreviated YAML snippet shown.

Video Reference: 1.2.3 YAML Parsing

7. What is the initial phase used when employing a Test-Driven Development methodology?

- A. Coding
- B. Planning
- C. Testing
- D. Debugging

Answer: C

Explanation: Contrary to some other methods, Test-Driven Development (TDD) begins with a testing phase. This results in shorter development cycles and a better overall code at the end. It also provides us the opportunity to create unit tests as we are developing, which may be reused against other code later.

Video Reference: 1.3.1 Test-Driven Development

8. What is the term for our general plan at the beginning of development when using the Agile Method?

- A. Sprint
- B. Product Backlog
- C. Cycle
- D. Scrum

Answer: B

Explanation: The Agile Method starts with a plan which is open to changes informed by feedback and review, creating a more general idea that we develop as we go along. This starting point is referred to as our Product Backlog, which includes details about all the features that need to be included in the product and what needs to be accomplished.

Video Reference: 1.3.2 Agile Method

9. What is the final step in the Lean Method before the cycle repeats?

- A. Seek perfection
- B. Create flow
- C. Map value stream
- D. Establish pull

Answer: A

Explanation: The team should always be looking for ways to implement improvements and increase efficiency, to bring the greatest value to the customer. This final step is referred to as "Seek Perfection." Realistically perfection can never be achieved, which is why the Lean Method is a cyclical model, encouraging constant improvement.

Video Reference: 1.3.3 Lean Method

10. Which of the following is a characteristic of the Waterfall Method?

- A. Customer feedback is timely and constant
- B. Value is created throughout the process
- C. Much more inexpensive than other methods
- D. Everything is planned at the beginning of a project

Answer: D

Explanation: The older Waterfall Method does not work well with our modern software development needs and challenges. Everything is planned at the beginning of a project, making the project harder to pivot and change when needed. Additional downsides include the facts that no value is created until the end of the process, and lack of timely feedback creates a challenge for ensuring quality.

Video Reference: 1.3.4 Waterfall Method

11. Which Python structure allows us to run the same piece of code multiple times in a program to repurpose the code?

- A. Classes
- B. Modules
- C. Functions
- D. Blocks

Answer: C

Explanation: By defining a block of code as a function, we can repurpose our code multiple times throughout a program, which saves us time as opposed to writing the same code repeatedly.

Video Reference: 1.4.1 Python Functions

12. Which Python structure is closely associated with objects?

- A. Modules
- B. Classes
- C. Functions
- D. Blocks

Answer: B

Explanation: Python objects are anything you want to manipulate or change while working through the code. Classes are closely associated with objects and are simply code templates for creating objects.

Video Reference: 1.4.2 Python Classes

13. Which Python structure allows us to import code from other external files into our main code?

- A. Blocks
- B. Functions
- C. Modules
- D. Classes

Answer: C

Explanation: Modules are another way we can save time when coding and decrease the overall size of our main code. By using an *import* statement, we can insert custom created modules or built in Python modules into our code from an external file.

Video Reference: 1.4.3 Python Modules

14. Which common design method leverages the Separation of Concerns (SoC) principle for designing modular code?

- A. Model-View-Controller (MVC)
- B. Observer
- C. Version Control
- D. Git

Answer: A

Explanation: Model-View-Controller (MVC) leverages the Separation of Concerns (SoC) principle. SoC is a design principle for separating our code into separate modular pieces. By doing this, we have more opportunity for upgrading software by changing the smaller components, rather than changing the larger overall code.

Video Reference: 1.5.1 Model-View-Controller (MVC)

15. Which design method is associated with one-to-many relationships, such as database design?

- A. Version Control
- B. Model-View-Controller (MVC)
- C. Git
- D. Observer

Answer: D

Explanation: The Observer pattern is used for one-to-many relationships, a common principle in database design. This means that a specific object is being accessed by multiple programs or individuals. This consists of a subject and one or more observers.

Video Reference: 1.5.2 Observer

16. Which principle means we are tracking all the changes made to a file or a set of files in a historical perspective?

- A. Observer

- B. Version Control
- C. Model-View-Controller (MVC)
- D. Logging

Answer: B

Explanation: Version control means we are tracking the changes made to a file or a set of files, checking the entire database of files where our code is stored. Software tools allow us to record the changes in a historical manner, such as the Git version control system. Typically, we use a central location as a repository for all of the data, where multiple entities can have working copies.

Video Reference: 1.5.3 Version Control

17. Which Git version control operation allows us to create a copy of a repository on our local file system?

- A. Commit
- B. Clone
- C. Push
- D. Merge

Answer: B

Explanation: The clone operation allows us to create a copy of a repository on our local file system. The command “git clone” followed by the URL for a repository will clone the repository contents into our current working directory on the system.

Video Reference: 1.6.1 Clone

18. Which Git version control operation allows us to track the status of a file including any changes made?

- A. Add/Remove
- B. Diff
- C. Branch
- D. Clone

Answer: A

Explanation: The add/remove operation is how we add or remove files from being tracked with Git version control. When we add files, any changes made to the file will be tracked for versioning purposes. The command “git add” followed by the file name will add a specific file for tracking. Optionally, we can add an entire directory and its contents by using the “git add -A” command.

Video Reference: 1.6.2 Add/Remove

19. Which Git version control operation moves a file from our index (or staging area) to our local repository?
- A. Merge
 - B. Push
 - C. Commit
 - D. Diff

Answer: C

Explanation: The commit operation moves a file from our staging area and commits the file changes to our local repository. This is how we sync our staging area to the local repository. The “git status” command will show any changes which are ready to be committed, and the “git commit” command is how we finalize the file commit.

Video Reference: 1.6.3 Commit

20. Which Git version control operation will add all tracked files from our local file system to a central remote repository such as GitHub?
- A. Commit
 - B. Push/Pull
 - C. Clone
 - D. Diff

Answer: B

Explanation: To truly leverage the full power of Git version control, a centrally accessible repository can be used to allow multiple developers to access the same code. To send local file changes to a remote repository, the “git push” command is used, along with the name of a configured remote repository that you have setup.

Video Reference: 1.6.4 Push/Pull

21. Which Git version control operation allows us a safe way to separate our development area from a main local repository containing known working code?
- A. Clone
 - B. Add/Remove
 - C. Diff

D. Branch

Answer: D

Explanation: Branches are a feature which allow us to separate our development area from a main local repository. This allows us a safe way to work on new features or updates. The command “git branch --list” will show us a list of our current local branches. We can create a new branch using the command “git branch” followed by the name we want to assign that branch.

Video Reference: 1.6.5 Branch

22. Which Git version control operation allows us to join the contents of a branch development area with a main repository locally?

- A. Diff
- B. Push/Pull
- C. Merge
- D. Add/Remove

Answer: C

Explanation: The merge operation allows us to join multiple branches together into a main coding repository. When working in the main repository, the “git merge” command followed by the name of the branch we want to add into this repository, will show any conflicts and attempt to merge the files.

Video Reference: 1.6.6 Merge

23. Which Git version control operation is used as a file comparison tool?

- A. Merge
- B. Branch
- C. Clone
- D. Diff

Answer: D

Explanation: The diff operation is a file comparison tool that will show us the difference in a set of files or development branches. That’s helpful when we’re merging branches or when multiple developers are working on the same files, so that we don’t unintentionally overwrite important code.

Video Reference: 1.6.7 Diff

24. Which method of authentication for REST API is typically limited only to operations where the READ function is required?

- A. Basic Authentication
- B. OAUTH Authentication
- C. Open Authentication
- D. API Key Authentication

Answer: D

Explanation: API Key Authentication uses pre-shared keys known by the client and the server. Because key transmission is susceptible to interception, its recommended to be used only in instances where the READ function is the only requirement. That helps limit the potential of a bad actor because if the key is intercepted, they don't have full access and permissions.

Video Reference: 2.1.1 Fundamentals of REST API

25. What type of HTTP messages are used by Webhooks?

- A. HTTP UPDATE
- B. HTTP READ
- C. HTTP POST
- D. HTTP DELETE

Answer: C

Explanation: Webhooks are HTTP POST messages which are triggered by a particular event in an automated manner. These are commonly referred to as Reverse APIs. Webhooks are a way we can get notification messages about an event that has occurred on a system without the need for constant polling.

Video Reference: 2.1.2 Webhooks

26. Which of the following is NOT a characteristic of a true RESTful API?

- A. Stateful
- B. Uniform Interface
- C. Cacheable or Non-cacheable Label
- D. Layered System

Answer: A

Explanation: True RESTful design has certain characteristics, or what we call constraints. All the listed options are constraints for REST APIs except for stateful. REST APIs should instead be

stateless, meaning that the server does not store information about the last action from a client. Every request made to the server should be treated as a new request without any history.

Video Reference: 2.1.3 REST Constraints

27. When using Python to make a REST API call, which built in library can be used in the script to perform a GET request?

- A. Requests
- B. Get
- C. Math
- D. XML

Answer: A

Explanation: The built in requests library in Python is an HTTP library that allows us to easily send HTTP requests. This can be used to make a GET request from Python.

Video Reference: 2.1.4 REST API Call with Python

28. Which range of HTTP response codes are used to indicate a client-side error?

- A. 2XX codes
- B. 3XX codes
- C. 4XX codes
- D. 5XX codes

Answer: C

Explanation: 400 range response codes are errors indicating a client-side error. This can include things like failed authorization or incorrect syntax from the request.

Video Reference: 2.2.1 Common HTTP Response Codes

29. In which part of an HTTP response would we see things like the host, user agent and cookie?

- A. URL
- B. Response code
- C. Arguments
- D. Headers

Answer: D

Explanation: The headers section is where we would see information such as the port and protocol being used, the host, the user agent, and the cookie, among other things.

Video Reference: 2.2.2 Parts of an HTTP Response

30. Which HTTP response code indicates a likely issue with incorrect authentication credentials?

- A. 500 Internal Server Error
- B. 403 Forbidden
- C. 401 Unauthorized
- D. 501 Not Implemented

Answer: C

Explanation: The 401 Unauthorized HTTP response code indicates that authentication credentials have not been provided or are incorrect. This is in contrast to a 403 Forbidden HTTP response code, where credentials are valid but the account does not have the required permission to access the resource.

Video Reference: 2.2.3 Troubleshooting Using HTTP Responses

31. Which API authentication mechanism is the least secure?

- A. OAUTH Authentication
- B. Basic Authentication
- C. Token Authentication
- D. API Key Authentication

Answer: B

Explanation: Basic authentication is the least secure method for API authentication, where credentials are passed in an unencrypted manner when using HTTP instead of HTTPS. Basic authentication uses a simple username and password for authentication.

Video Reference: 2.3.1 Basic Authentication

32. Which API authentication mechanism allows us to enter our username and password a single time to generate an encrypted value for accessing resources?

- A. Token Authentication
- B. API Key Authentication
- C. Basic Authentication
- D. Secure Authentication

Answer: A

Explanation: Token authentication allows us to use a valid username and password combination to retrieve a valid encrypted token for access to resource access. This method means we don't have to continuously enter the login credentials or pass those in an unsecure manner inside of the request we are making. Tokens can have limited time-based access which will provide expiring access rather than indefinite access.

Video Reference: 2.3.2 Token Authentication

33. Which API authentication mechanism is primarily used for overall project access rather than at the individual user level?

- A. General Authentication
- B. Encrypted Authentication
- C. Token Authentication
- D. API Key Authentication

Answer: D

Explanation: API key authentication is primarily used for overall project access. This is in contrast with token authentication, which is primarily utilized at the individual user level.

Video Reference: 2.3.3 API Key Authentication

34. Which subset of common HTTP methods are used by Remote Procedure Calls (RPCs)?

- A. POST and PUT
- B. PATCH and DELETE
- C. GET and POST
- D. PUT and GET

Answer: C

Explanation: As opposed to REST APIs which use GET, POST, PUT, PATCH, and DELETE HTTP methods, Remote Procedure Calls (RPCs) use only the GET and POST methods. For this reason, RPCs are action-oriented, where we're concerned with executing code on a remote device.

Video Reference: 2.4.1 Remote Procedure Call (RPC)

35. Which style of API does not use real-time communication?

- A. Synchronous

- B. Asynchronous
- C. One-way
- D. Delayed

Answer: B

Explanation: Asynchronous APIs do not communicate in real-time, as opposed to synchronous APIs which feature scheduled real-time communication and interaction. Email communication is an illustration of this, where there is no need to instantly respond within a timeframe and responses can be delayed.

Video Reference: 2.4.2 Synchronous vs Asynchronous

36. What is the common name for a set of tools, libraries and documentation that allows us to interact with a REST API?

- A. Framework
- B. Program
- C. Script
- D. SDK

Answer: D

Explanation: A software development kit (SDK) is a set of tools, libraries and documentation provided by the manufacturer of a hardware platform, operating system, or programming language. This will provide the necessary components for interacting with a system.

Video Reference: 3.1.1 Python Script with Cisco SDK

37. Which of the following is appropriate for obtaining a list of the organizations available through the GET method when using Postman to interact with the Cisco Meraki API?

- A. {{baseUrl}}/organizations/:organizationId
- B. {{baseUrl}}/organizations
- C. {{baseUrl}}/organizations/:organizationId/networks
- D. {{baseUrl}}/organizations/:organizationId/devices

Answer: B

Explanation: When using the GET method to perform a request against the Meraki API using Postman, all the above options are valid. However, if we simply want to see a list of the organizations associated with a Meraki controller, setting the get method to {{baseUrl}}/organizations will perform this action.

Video Reference: 3.2.1 Meraki

38. Which type of API authentication method is used by Cisco DNA Center?

- A. Token Authentication
- B. Basic Authentication
- C. API Key Authentication
- D. Open Authentication

Answer: A

Explanation: Cisco DNA Center uses token authentication, where we need to retrieve a token to obtain access to information. Using the Cisco DNAC Sandbox collection provides a pre-configured POST method that can be used to retrieve a token, which we can save in our Postman environment for interaction.

Video Reference: 3.2.2 DNA Center

39. Which type of API authentication method is used by Cisco APIC?

- A. Open Authentication
- B. API Key Authentication
- C. Basic Authentication
- D. Token Authentication

Answer: D

Explanation: Just as with Cisco DNA Center, Cisco APIC requires a token to interact through the API. This requires sending a POST method with the basic username and password, requesting a validation token from APIC. Performing methods without this token will result in a 403 Forbidden HTTP response code.

Video Reference: 3.2.3 Application Centric Infrastructure

40. Which area in Postman allows you to create preconfigured areas to be used with specific devices, which can hold key:value pairs for variables like the base URL, the token, the username, and the password?

- A. Environments
- B. Flows
- C. Collections
- D. Requests

Answer: A

Explanation: A Postman environment is a set of variables that can be used for requests. An environment can store key: value pairs for specific devices, which can be referenced in our requests using variables, which saves us from entering the same information multiple times with our methods. Common information stored in an environment include the device base URL, the token, the username, and the password.

Video Reference: 3.2.4 SD-WAN

41. With Cisco Network Services Orchestrator, what are required to translate between XML format and our network configuration?

- A. YAML
- B. JSON
- C. NEDS
- D. LIB

Answer: C

Explanation: Network Element Drivers (NEDS) are necessary to translate between XML data and a valid network configuration. Cisco NSO provides NEDS for various versions of the Cisco CLI since many devices use different commands. Some of those include NEDS for Cisco ASA, Cisco IOS-XR, and Cisco IOS-NX.

Video Reference: 3.2.5 Network Services Orchestrator

42. Within a Cisco UCS-based network, what type of device is used to interconnect all UCS-capable servers into a single point of management?

- A. UCS Fabric Extender
- B. UCS Series Servers
- C. UCS Fabric Interconnects
- D. UCS Manager

Answer: C

Explanation: When using Cisco UCS Manager, UCS Fabric Interconnects are used to integrate our devices into a single fabric of connectivity for management. As a best practice, UCS Fabric Interconnects are deployed as a pair for failover and high availability.

Video Reference: 3.3.1 UCS Manager

43. Which component used within Cisco UCS Director are the largest units of information?

- A. Workflows
- B. Tasks
- C. Volumes
- D. Libraries

Answer: D

Explanation: Libraries are the largest unit within Cisco UCS Director, containing collections of workflows or pre-defined tasks. Tasks are the smallest unit, representing a single action. Multiple tasks are organized into workflows as collections of tasks, and multiple workflows can be contained within a library.

Video Reference: 3.3.2 UCS Director

44. Which area within Cisco Intersight shows all the devices under our control which we have claimed?

- A. Assets
- B. Profiles
- C. Targets
- D. Policies

Answer: C

Explanation: From the Cisco Intersight management portal, under the admin menu we can choose the Targets area to see all of the claimed devices we have under our control.

Video Reference: 3.3.3 Intersight

45. Which method is required for creating a new room in Cisco Webex through API interaction?

- A. POST
- B. PUT
- C. GET
- D. UPDATE

Answer: A

Explanation: The POST method is used for creating a new room through the Cisco Webex API. The GET method would point to <https://webexapis.com/v1/rooms> and would require a title for the room to create the new space.

Video Reference: 3.4.1 Webex Teams

46. Which API is used by Cisco collaboration endpoint software such as Cisco CE and Cisco RoomOS?

- A. ceAPI
- B. cAPI
- C. osAPI
- D. xAPI

Answer: D

Explanation: The xAPI allows developers to programmatically invoke commands and retrieve the status of devices that run Cisco CE (collaboration endpoint) software and Webex RoomOS. This xAPI allows for bi-directional communication with third-party applications and control systems.

Video Reference: 3.4.2 Webex Devices

47. Which type of API interface is used for Cisco CUCM interaction?

- A. Open API
- B. xAPI
- C. AXL API
- D. SDK API

Answer: C

Explanation: The Administrative XML Web Service (AXL) is an XML-based interface that provides a mechanism for inserting, retrieving, updating, and removing data from the Unified Communication configuration database. Developers can use AXL to CREATE, READ, UPDATE, and DELETE objects such as gateways, users, devices, route-patterns and more.

Video Reference: 3.4.3 Cisco UCM

48. Which Cisco collaboration solution provides an agent and supervisor desktop that we can affect through REST API interaction?

- A. Finesse
- B. Cisco UCM
- C. Cisco Webex
- D. Cisco UCS

Answer: A

Explanation: Cisco Finesse is an agent and supervisor desktop that integrates traditional contact center functions into a thin-client desktop. Finesse includes a robust API for creating custom applications.

Video Reference: 3.4.4 Finesse

49. Which key:value pair used to identify the authentication token with the Firepower API?

- A. X-auth-token:<authentication token value>
- B. X-auth-access-token:<authentication token value>
- C. X-auth-bearer-token:<authentication token value>
- D. X-auth-secure-token:<authentication token value>

Answer: B

Explanation: The appropriate key:value pair for identifying the authentication token when interacting with the Firepower API is X-auth-access-token:<authentication token value>, where the actual token value is placed as the value for this key. Additionally, Firepower requires a key:value pair for the refresh token, as outlined by the API documentation.

Video Reference: 3.5.1 Firepower

50. Which cloud-based Cisco platform is used as a secure internet gateway for things like malware prevention and secure DNS routing?

- A. AMP
- B. Umbrella
- C. ISE
- D. ThreatGrid

Answer: B

Explanation: Cisco Umbrella acts as a secure internet gateway and provides DNS-layer security. This allows us to route endpoints over secure DNS destinations, prevent malware and ransomware, and more.

Video Reference: 3.5.2 Umbrella

51. When interacting with Cisco Secure Endpoint (formerly Cisco AMP) through API, what type of encoding should we use for the client ID and API key for use within Postman?

- A. Unicode
- B. ASCII

- C. Base64
- D. XML

Answer: C

Explanation: The client ID and API key values need to be converted to Base64 encoding, which is a well-known binary-to-text encoding scheme. This can be performed natively in a Linux terminal by using the command “echo <client ID>:<API Key> | base64”, where the actual client ID and API Keys are used as a key:value pair.

Video Reference: 3.5.3 AMP

52. Which appropriate key:value pair will set our content type to be formatted in JSON when interacting with the Cisco ISE API?

- A. Content-type:application/json
- B. Content:type/json
- C. Content:json
- D. Content-type:json

Answer: D

Explanation: Adding the key:value pair Content-type:application/json is how we declare the content type in the request header for Cisco ISE. Without explicitly declaring the content type of a resource, the client may attempt to automatically detect the type, but the result may not be accurate.

Video Reference: 3.5.4 ISE

53. Which type of malware analysis used by Cisco Secure Malware Analytics (formerly Cisco ThreatGrid) is performed in an isolated virtual environment?

- A. Dynamic malware analysis
- B. Static malware analysis
- C. Virtual malware analysis
- D. Isolated malware analysis

Answer: A

Explanation: The two types of analysis used with this solution are static and dynamic malware analysis. Static analysis compares the cryptographic hash of files to known malware signatures, examining the code without executing anything. Dynamic analysis runs the code and observes the behavior in an isolated virtual environment.

Video Reference: 3.5.5 ThreatGrid

54. Which command within the Cisco NX-OS CLI line will enable the developer sandbox?

- A. feature nsox
- B. feature sandbox
- C. feature nxapi
- D. feature devbox

Answer: C

Explanation: From global configuration mode, the command “feature nxapi” will enable the developer sandbox option. This is a webform hosted directly on the switch used to translate NX-OS CLI commands into their equivalent XML or JSON payloads. It will also convert NX API REST payloads into their CLI equivalents.

Video Reference: 3.6.1 NX-OS API

55. Which DevNet resource provides a way for us to instantly access Cisco devices through a web browser?

- A. API Documentation
- B. Learning Labs
- C. Code Exchange
- D. Sandbox

Answer: D

Explanation: The DevNet Sandbox provides both always-on and reservation devices from various Cisco device categories. This includes Networking, Data Center, Cloud, Collaboration, and Security, among others.

Video Reference: 3.7.1 Sandbox

56. Which DevNet resource provides an online, publicly available repository of code related to Cisco technologies?

- A. API Documentation
- B. Learning Labs
- C. Code Exchange
- D. Sandbox

Answer: C

Explanation: The Cisco Code exchange provides a free repository of curated code for various Cisco technologies. This portal will allow you to explore their various GitHub repositories and projects for code related to your specific needs or solution.

Video Reference: 3.7.2 Code Exchange

57. What is the recommended way to obtain in-depth assistance with topics related to DevOps and Cisco DevNet principles from the DevNet Developer Support site?

- A. Knowledge Base
- B. Community Forum
- C. Chat
- D. Support Ticket

Answer: B

Explanation: The option for 1-on-1 support tickets ended in April 2022, so the new recommendation is to leverage the Community Forum. Here you can engage with DevNet and community experts on technology-specific questions.

58. What is the term for attributes outlined such as integers or strings within the YANG data model?

- A. Key
- B. Namespace
- C. Leaf
- D. List

Answer: C

Explanation: A leaf is an attribute within the YANG data model that describes the type of data. That can include things like integer, string, Boolean, and more. For example, if we have an interface description described in the YANG data model, this will have the leaf “string”, because the interface description would be a string text value.

Video Reference: 3.8.1 YANG Data Model

59. When a router receives a NETCONF request from another device, what is the term for this device within the context of NETCONF?

- A. Syslog
- B. Trap
- C. Manager

D. Agent

Answer: D

Explanation: NETCONF works in a client-server manner. The client is referred to as a Manager in the context of NETCONF, which is running an application that makes a request to another device such as a router. The router receiving the request is referred to as an Agent.

Video Reference: 3.8.2 NETCONF

60. Which protocol is used by RESTCONF for sending and receiving information?

- A. SNMP
- B. SSH
- C. HTTPS
- D. DNS

Answer: C

Explanation: RESTCONF is like having a client pointing to a secure web server. Similarly, RESTCONF uses the HTTPS protocol to send and retrieve information from a device using HTTP verbs. This supports the same YANG data model definitions used by NETCONF, and supports both JSON and XML formatting.

Video Reference: 3.8.3 RESTCONF

61. Which type of cloud service model provides users with quick access to cloud-based web applications?

- A. SaaS
- B. PaaS
- C. IaaS
- D. DaaS

Answer: A

Explanation: Software as a Service (SaaS) is a cloud service model that is used to provide clients with quick access to cloud-based web applications. There are no downloads or installation required on the client side. Common examples include Dropbox and Office 365.

Video Reference: 4.1.1 Cloud Service Models

62. Which type of cloud deployment model is provisioned for open use?

- A. Private Cloud
- B. Community Cloud
- C. Public Cloud
- D. Hybrid Cloud

Answer: C

Explanation: Public Cloud deployments are provisioned for open public use and is the most common type of cloud deployment mode. Common examples include AWS, Azure, and Dropbox.

Video Reference: 4.1.2 Cloud Deployment Models

63. Within the NIST Cloud Deployment Reference Architecture, under which role would we categorize end users of a cloud solution?

- A. Cloud Provider
- B. Cloud Broker
- C. Cloud Consumer
- D. Cloud Carrier

Answer: C

Explanation: As outlined by NIST SP 500-292, the end users are found under the Cloud Consumer category. This entity includes any customer using services provided by a Cloud Provider.

Video Reference: 4.1.3 NIST Cloud Reference Architecture

64. Which of the following is NOT a characteristic of Edge Computing?

- A. Better security and privacy
- B. Slightly higher cost associated
- C. Lower latency for real-time data
- D. Balance between local and cloud resources

Answer: B

Explanation: All the outlined options are characteristics of Edge Computing, except for higher associated costs. Edge Computing actually offers cost savings compared to the requirement of having powerful, centralized processing.

Video Reference: 4.1.4 Edge Computing

65. Which type of Hypervisor runs directly on a server's hardware?

- A. Type-1
- B. Type-2
- C. Type-3
- D. Type-4

Answer: A

Explanation: Hypervisors can be categorized as either Type-1 or Type-2. Type-1 Hypervisors are also known as Native or Bare Metal Hypervisors, where the software runs directly on the server's hardware. Type-2 Hypervisors are known as Hosted Hypervisors, where they run in a traditional operating system as an additional layer.

Video Reference: 4.2.1 Hypervisors

66. Which of the following is NOT a characteristic of Containers?

- A. Run on a single host OS
- B. Smaller than VMs
- C. Consist of packages of applications, software, and libraries
- D. More resource intensive than VMs

Answer: D

Explanation: All the options listed are characteristics of Containers, except for being more resource intensive than Virtual Machines (VMs). Because containers run on a single host OS with shared resources, this means Containers require less resources than VMs and are more efficient.

Video Reference: 4.2.3 Containers

67. Under which cloud service model would we categorize Serverless Computing?

- A. IaaS
- B. SaaS
- C. FaaS
- D. PaaS

Answer: C

Explanation: Serverless Computing introduces an additional cloud service model, in addition to SaaS, PaaS, and IaaS. This is called Function as a Service (FaaS). FaaS will execute code on-demand with no permanent resources necessary.

Video Reference: 4.2.4 Serverless Computing

68. Which step within the CI/CD pipeline involves staging and deploying the application?

- A. Source
- B. Deploy
- C. Build
- D. Test

Answer: B

Explanation: The deploy step is the final step of the CI/CD pipeline, where software is staged and deployed into production. This typically happens in multiple deployment environments, which would include an internal beta testing area and a production environment.

Video Reference: 4.3.1 DevSecOps Principles

69. After writing a Unit Test, what is next second step which will inform the remainder of our Test-Drive Development cycle?

- A. Testing fails
- B. Refactoring
- C. Testing passes
- D. Merging code

Answer: A

Explanation: Step 2 in the Test-Driven Development cycle is having a Unit Test fail. Initially, we write a test with failure in mind. The failure of the Unit Test helps inform us about how we should write code to pass the test and achieve the desired results.

Video Reference: 4.3.2 Python Unit Test

70. What is the term for a text document containing commands used for assembling images in Docker?

- A. Dockercontainer
- B. Dockerfile
- C. Dockerimage
- D. Dockerscript

Answer: B

Explanation: At its core, a Dockerfile is essentially a script containing instructions. These instructions are used for assembling images in Docker, which in turn can be used to build containers.

Video Reference: 4.3.4 Dockerfiles

71. At which layer of the Vulnerability Stack for threats would we find web applications?

- A. Network
- B. Operating system
- C. Databases
- D. Third-party components

Answer: D

Explanation: The layer for third-party components is where we would categorize our web applications in the Vulnerability Stack for web applications and web servers.

Video Reference: 4.3.5 Vulnerability Stack

72. According to the most recent OWASP Top 10 Web Application Threats list, what is rated as the number one danger to our web applications?

- A. Injection
- B. Cryptographic failures
- C. Broken access control
- D. Insecure design

Answer: C

Explanation: The number one rated threat to our web applications is broken access control, as identified by the Open Web Application Security Project (OWASP). This relates to what a user can perform on a network or a device, and failures often lead to unauthorized data exposure. We want to make sure that users only have the amount of access necessary for performing their job duties, and no more.

Video Reference: 4.3.6 Web Application Threats

73. Which of the following is not an advantage of Model-Driven Programmability for infrastructure automation?

- A. Does not require vendor-specific syntax
- B. Unified way of interacting

- C. Complete control over devices
- D. Reliable and repeatable management and configuration

Answer: C

Explanation: All the options outlined are advantages of infrastructure automation, except for having complete control over devices. Although infrastructure automation is very reliable and saves time, there may be cases where you still need to access a device over the CLI for more powerful management or configuration capabilities.

Video Reference: 5.1.1 Infrastructure Automation

74. With controller-level management, which type of interface is used for integrating third-party systems through REST APIs?

- A. Northbound
- B. Southbound
- C. Eastbound
- D. Westbound

Answer: A

Explanation: The Northbound interface is where we could find our REST API interface, where we can integrate third-party systems to connect to and manage the network through programming. The Southbound interfaces connect out to network devices for management, typically using CLI-based interactions over Telnet, SSH, NETCONF, RESTCONF, or SNMP.

Video Reference: 5.2.1 Controller-level Management

75. Which type of network management approach features a single central point of administration?

- A. Central-level management
- B. Server-level management
- C. Device-level management
- D. Controller-level management

Answer: D

Explanation: Controller-level management features a single centralized point for management and configuration. An example of this from the Cisco realm is DNA Center. This is in contrast with device-level management, which requires individual interaction with each device.

Video Reference: 5.2.2 Device-level Management

76. What is the name of the YAML file in pyATS where we define devices to run tests against?

- A. Inventory.yaml
- B. Assets.yaml
- C. Testbed.yaml
- D. Devices.yaml

Answer: C

Explanation: The testbed file is a YAML formatted document containing definitions for various objects that are in our environment, such as routers, switches, servers and more. In order to run tests, we need to properly define our testbed file with devices to test against.

Video Reference: 5.3.2 pyATS

77. Which approach to Infrastructure as Code (IaC) involves defining what the desired end state of the system should look like?

- A. Manual approach
- B. Defining approach
- C. Imperative approach
- D. Declarative approach

Answer: D

Explanation: The declarative approach in IaC is where we define what the desired end state of the system should look like, and then allow the automation system to execute all of the steps that need to happen in order for the system to reach this state.

Video Reference: 5.4.2 Infrastructure as Code

78. Which well-known automation tool has structures such as Recipes and Cookbooks?

- A. Chef
- B. Puppet
- C. Ansible
- D. Python

Answer: A

Explanation: Chef uses a pull model with a central server for automation and orchestration. A Recipe is written in the Ruby programming language and are contained in Cookbooks. Cookbooks are simply collections of recipes and other files used to facilitate automation.

Video Reference: 5.4.3 Automation Tools

79. What is the term used with Puppet automation for a description of a desired machine state?

- A. Module
- B. Manifest
- C. Resource
- D. Collection

Answer: C

Explanation: Resources are structures within Puppet Catalogs that contain a description of the desired state of the system. This can include things like whether a certain file or user should exist, or if a specific package should be installed. These are the smallest building blocks of the language and represent a single element.

Video Reference: 5.4.3 Automation Tools

80. Which Ansible structure contains a list of nodes which are under the management of the Ansible Controller?

- A. Playbook
- B. Inventory File
- C. Module
- D. Play

Answer: B

Explanation: The Ansible Inventory File contains a list of nodes which are under the management of the Ansible system. This can include servers, switches, and other devices. Playbooks containing tasks for execution will reference the Inventory File for information about how to reach a device.

Video Reference: 5.4.3 Automation Tools

81. When interpreting a YANG data model, what indicates that a particular field is configurable, as opposed to only being able to retrieve information about the field?

- A. ro
- B. rw
- C. conf
- D. string

Answer: B

Explanation: The rw designation (read/write) indicates that we can configure a particular field programmatically. This is opposed to the ro designation (read-only), which means we can only retrieve information about that field but not make changes.

Video Reference: 5.5.1 Interpreting YANG Models

82. What port is used for NETCONF by Cisco IOS-XE?

- A. Port 443
- B. Port 9339
- C. Port 50052
- D. Port 830

Answer: D

Explanation: Chef uses a pull model with a central server for automation and orchestration. A Recipe is written in the Ruby programming language and are contained in Cookbooks. Cookbooks are simply collections of recipes and other files used to facilitate automation.

Video Reference: 5.5.2 Interpreting a NETCONF Query

83. What port is used for NETCONF by Cisco IOS-XE?

- A. Port 443
- B. Port 9339
- C. Port 50052
- D. Port 830

Answer: A

Explanation: In Cisco IOS-XE, RESTCONF listens on any IP address assigned to the system. RESTCONF operates on port 443 when enabled and uses HTTPS for transport.

Video Reference: 5.5.3 Interpreting a RESTCONF Query

84. You have two Git branches on your local system, one named “main” and the other named “newfeatures.” When working in the NewFeatures branch, which terminal command will allow you to see the differences between these branches and files?

- A. git diff main
- B. git diff newfeatures

- C. git diff branch
- D. git diff branch –list

Answer: A

Explanation: The terminal command “git diff main” while working in the branch named “newfeatures” will allow you to see the differences between the branches and files.

Video Reference: 5.5.4 Interpreting a Unified Diff

85. Which of the following is NOT a best practice recommendation for code review?

- A. Review less than 400 lines at a time
- B. Take breaks throughout code review
- C. Review all code in one sitting
- D. Review at a pace no faster than 400 lines per hour

Answer: C

Explanation: SmartBear, a company used by Cisco for their internal code review processes, outlines several best practices for code review. They include all the above recommendations, except for reviewing the code all in one sitting. It’s more productive to take breaks to avoid overlooking code errors due to fatigue.

Video Reference: 5.6.1 Code Review

86. Which of the following is NOT a characteristic of Sequence Diagrams?

- A. Used to illustrate REST APIs visually
- B. Provide the necessary API keys for interaction
- C. Common method for developer teams
- D. Helps outline project goals and requirements

Answer: B

Explanation: Sequence Diagrams are used as high-level flowcharts that outline API interaction in a visual manner. These are commonly used with developer teams to outline project goals and requirements. Being a high-level overview, they would not contain detailed information such as the actual API keys.

Video Reference: 5.6.2 Sequence Diagrams

87. What is the size of a MAC address?

- A. 64 bits

- B. 32 bits
- C. 128 bits
- D. 48 bits

Answer: D

Explanation: A MAC address is a 48-bit identifier burned into the hardware of a device. The first 24 bits are dedicated to the Organizationally Unique Identifier (OUI), while the remaining 24 bits are assigned by the vendor.

Video Reference: 6.1.1 MAC Addresses

88. Which interface mode is used to carry traffic for multiple designated VLANs?

- A. VLAN mode
- B. Access mode
- C. Trunk mode
- D. Multi mode

Answer: C

Explanation: Interfaces designated with trunk mode are not assigned to a specific VLAN and are used to carry traffic from multiple VLANs. Trunks are what we use to interconnect switches to other switches or routers. These interfaces are often referred to as tagged ports.

Video Reference: 6.1.2 VLANs

89. What is the size of an IPv4 address?

- A. 64 bits
- B. 32 bits
- C. 128 bits
- D. 48 bits

Answer: B

Explanation: An IPv4 address is a 32-bit number, containing four sections (called octets) separated by decimals. These octets are 8 bits each, for a total of 32 bits.

Video Reference: 6.1.3 IP Addresses

90. Which address structure is used by a router for sending traffic to its next destination?

- A. IP address

- B. MAC address
- C. DNS address
- D. HTTP address

Answer: A

Explanation: Routers work at Layer 3 and are primarily concerned with IP addressing for traffic forwarding. This is in contrast with a typical Layer 2 switch, which uses MAC addressing for traffic forwarding.

Video Reference: 6.1.4 Routing

91. At which layer of the OSI Model do we find the packets?

- A. Layer 1
- B. Layer 2
- C. Layer 3
- D. Layer 4

Answer: C

Explanation: Each layer of the OSI Model contains one or more protocol data units (PDUs). At Layer 3 is where we find packets as the PDU. Layer 1 contains bits, Layer 2 contains frames, Layer 4 contains segments and datagrams, and Layers 5-7 have the general data PDU.

Video Reference: 6.1.6 OSI Model

92. At which layer of the OSI model do we categorize a typical switch?

- A. Layer 1
- B. Layer 2
- C. Layer 3
- D. Layer 4

Answer: B

Explanation: Typically, we think of switches as operating at Layer 2 of the OSI Model, being concerned with MAC addressing for traffic forwarding. Although there are multi-layer switches that can operate at Layer 2 and Layer 3, traditionally we consider a switch a Layer 2 device.

Video Reference: 6.2.1 Switches

93. What type of device is necessary for routing traffic between different VLANs?

- A. Layer 4 device
- B. Layer 2 device
- C. Layer 1 device
- D. Layer 3 device

Answer: D

Explanation: VLANs are created on switches and require a Layer 3 device for routing between these virtual networks. Most commonly, we use a Layer 3 router to achieve this.

Video Reference: 6.2.2 Routers

94. What is the term for the firewall zone connecting out to the public Internet?

- A. Outside
- B. Inside
- C. DMZ
- D. Demarcation

Answer: A

Explanation: The Outside zone is also referred to as the Untrusted zone. This is the firewall zone we designate for the interface connecting our network out to the public Internet.

Video Reference: 6.2.3 Firewalls

95. Which network device is ideally suited for providing redundancy and low latency?

- A. Load balancer
- B. Firewall
- C. Virtual server
- D. Next-generation router

Answer: A

Explanation: Load balancers are a way that can provide redundancy in the network and ensure that we have low latency. These can be used to balance traffic out to multiple devices such as virtual servers, or even multiple redundant network connections.

Video Reference: 6.2.4 Load Balancers

96. Which plane of operation in networking is where we find user traffic transiting the router, not destined to, or initiated by the router itself?

- A. Data Plane
- B. Management Plane
- C. Control Plane
- D. Traffic Plane

Answer: A

Explanation: The Data Plane is often referred to as the User Plane or the Forwarding Plane. This plane determines what to do with inbound packets arriving, since they are not destined to the router itself.

Video Reference: 6.3.1 Data Plane

97. Which plane of operation in networking is considered to have its own ingress and egress ports?

- A. Data Plane
- B. Management Plane
- C. Control Plane
- D. Traffic Plane

Answer: C

Explanation: The Control Plane is considered as a separate entity, having its own ingress and egress ports. This allows for traffic filtering and rate limiting through Control Plane Policing (CoPP).

Video Reference: 6.3.2 Control Plane

98. Which plane of operation in networking is where we would see protocols like SSH and Telnet being used?

- A. Data Plane
- B. Management Plane
- C. Control Plane
- D. Traffic Plane

Answer: B

Explanation: The Management Plane is where we find traffic for the management of a device through its connection to the network. This would include protocols such as SNMP, Telnet, SSH, and FTP, among others.

Video Reference: 6

99. Which step during the DHCP exchange uses a broadcast message to locate a DHCP server?

- A. Offer
- B. Request
- C. Discover
- D. Acknowledgement

Answer: C

Explanation: The first step in the DHCP exchange involves a client sending out a Discover broadcast, to discover the presence of a DHCP server in the network. This is followed in succession by the Offer, Request, and Acknowledgement message exchanges.

Video Reference: 6.4.1 DHCP

100. Which well-known port is used by DNS?

- A. 80
- B. 1400
- C. 53
- D. 443

Answer: C

Explanation: Domain Name System (DNS) uses well-known port 53 for communication.

Video Reference: 6.4.2 DNS

101. When using Network Address Translation, what is the term for addresses found inside the NAT-enabled network?

- A. Inside Local
- B. Inside Global
- C. Outside Local
- D. Outside Global

Answer: A

Explanation: The term “inside” means our addresses are inside the NAT-enabled router, on our own network. The term “local” means the address is only valid on this local network, and not routable on the public Internet. Therefore, addresses found inside the NAT-enabled network are referred to as Inside Local addresses.

Video Reference: 6.4.3 NAT

102. Which version of SNMP added additional security features for encryption, integrity checking, and authentication services?

- A. SNMPv1
- B. SNMPv2
- C. SNMPv2c
- D. SNMPv3

Answer: D

Explanation: SNMPv3 is the recommended version to be used on modern networks, with included encryption, integrity checking, and user authentication. This also leverages a unique value called the Engine ID, which can only be used by a single SNMP entity on the domain.

Video Reference: 6.4.4 SNMP

103. Which well-known port is used by Network Time Protocol (NTP)?

- A. TCP 123
- B. UDP 123
- C. UDP 189
- D. TCP 189

Answer: B

Explanation: Network Time Protocol (NTP) uses well-known port 123 over UDP for communication. This is used to provide accurate and synchronized time throughout a network for log correlation.

Video Reference: 6.4.5 NTP

104. What is the second step in the TCP 3-way handshake process?

- A. SYN-ACK
- B. SYN
- C. ACK
- D. ACK-SYN

Answer: A

Explanation: The first step in the TCP 3-way handshake is a synchronization (SYN) message sent from a client requesting a TCP session with another device. Next, the remote device responds with a synchronization and acknowledgement (SYN-ACK) message, agreeing to start a TCP

session. The original client will then conclude the 3-way handshake with a final acknowledgement (ACK) message sent to the remote device.

Video Reference: 6.5.1 TCP and UDP

105. What range of ports is referred to as the well-known port range?

- A. 49151 - 65535
- B. 1024 - 49151
- C. 0 – 1023
- D. 500 – 1024

Answer: C

Explanation: The well-known port range consists of port numbers 0 through 1023. This is where we commonly see network services such as DNS, DHCP, HTTP and HTTPS. The registered port range is 1024 through 49151, while the ephemeral port range is 49152 through 65535.

Video Reference: 6.4.2 Ports and Protocols

106. At which OSI Model layer would we perform troubleshooting if we have a misconfigured or disabled switch port?

- A. Network Layer
- B. Data Link Layer
- C. Physical Layer
- D. Transport Layer

Answer: C

Explanation: Common troubleshooting steps for the Physical Layer include checking for defective cabling and wall ports, and ensuring general connections have no issues. This would also encompass checking for disabled or misconfigured switch ports.

Video Reference: 6.5.1 Physical Layer Issues

107. At which OSI Model layer are we performing troubleshooting when using Cisco IOS commands such as “show mac address-table”?

- A. Network Layer
- B. Data Link Layer
- C. Physical Layer
- D. Transport Layer

Answer: B

Explanation: The Data Link layer is concerned with physical MAC addressing. Commands such as “show mac address-table” in Cisco IOS can help identify places where there are problems with devices having the ability to learn MAC addresses.

Video Reference: 6.5.2 Data Link Layer Issues

108. At which OSI Model layer are we performing troubleshooting when using the ping command utility?

- A. Network Layer
- B. Data Link Layer
- C. Physical Layer
- D. Transport Layer

Answer: A

Explanation: The Network Layer is concerned with IP addressing and troubleshooting at this layer involves ensuring there is IP reachability and connectivity. The ping command utility is a common way we can do this on a network.

Video Reference: 6.5.3 Network Layer Issues

109. At which OSI Model layer are we performing troubleshooting when checking firewall access control lists for issues?

- A. Network Layer
- B. Data Link Layer
- C. Physical Layer
- D. Transport Layer

Answer: D

Explanation: Transport Layer issues deal with TCP and UDP port access problems. For instance, HTTP (port 80) or HTTPS (port 443) communication. Firewall access control lists are one structure that can cause potential problems at the Transport Layer, particularly if misconfigured.

Video Reference: 6.5.4 Transport Layer Issues

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Take good care,

Kevin Wallace, CCIEx2 #7945 Emeritus (Enterprise Infrastructure & Collaboration)